Open the document **evidence.doc**.

Make sure that your name, centre number and candidate number will appear on every page of this document. This document will contain your answers to each question.

Save this evidence document in your work area as:

evidence_ followed by your centre number_candidate number, for example: evidence_zz999_9999

A class declaration can be used to declare a record.

A list is an alternative to an array.

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2 A program stores the following ten integers in a 1D array with the identifier `arrayData`.

10 5 6 7 1 12 13 15 21 8

(a) Write program code for a **new program** to:

- declare the global 1D array, `arrayData`, with ten elements
- initialise `arrayData` in the main program using the data values shown.

Save your program as **question2**.

Copy and paste the program code into **part 2(a)** in the evidence document.

[2]

(b) (i) A function, `linearSearch()`, takes an integer as a parameter and performs a linear search on `arrayData` to find the parameter value. It returns `True` if it was found and `False` if it was not found.

Write program code for the function `linearSearch()`.

Save your program.

Copy and paste the program code into **part 2(b)(i)** in the evidence document.

[6]

(ii) Edit the main program to:

- allow the user to input an integer value
- pass the value to `linearSearch()` as the parameter
- output an appropriate message to tell the user whether the search value was found or not.

Save your program.

Copy and paste the program code into **part 2(b)(ii)** in the evidence document.

[4]

(iii) Test your program with one value that is in the array and one value that is not in the array.

Take a screenshot to show the result of each test.

Save your program.

Copy and paste the screenshots into **part 2(b)(iii)** in the evidence document.

[2]

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(c) The following bubble sort pseudocode algorithm sorts the data in `theArray` into descending numerical order. There are **five** incomplete statements.

```
PROCEDURE bubbleSort()

   DECLARE temp : INTEGER

   FOR x ← 0 to ……

        FOR y ← 0 to ……

            IF theArray[y] …… theArray[y + 1] THEN

               temp ← theArray[y]

               theArray[y] ← ……

               theArray[y + 1] ← ……

            ENDIF

        NEXT y

   NEXT x

ENDPROCEDURE
```

Write program code for the procedure `bubbleSort()` to sort the data in `arrayData` into descending order.

Save your program. Copy and paste the program code into **part 2(c)** in the evidence document.

[6]